

ECON 3201 Assignment 1

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Q1

My personal ECON 3201 Quarto website can be accessed using the following link:

Website: <https://firas7541-dot.github.io/ECON3201/>

The website contains a Home page with my name, credentials, and an About Me section, as well as a Blog page for my ECON 3201 assignments.

Q2

Let $X = \{1, 2, 3, \dots, 12\}$ be uniformly distributed with

$$f_X(x) = \frac{1}{12}.$$

(a)

For a valid probability distribution, all probabilities must be non-negative and their sum must equal 1.

$$\sum_{x=1}^{12} f_X(x) = \sum_{x=1}^{12} \frac{1}{12} = 12 \left(\frac{1}{12} \right) = 1.$$

Also,

$$f_X(x) = \frac{1}{12} \geq 0.$$

Therefore, $f_X(x)$ is a valid probability distribution.

(b)

The expected value is

$$E(X) = \sum_{x=1}^{12} x f_X(x).$$

Therefore,

$$E(X) = \frac{1}{12}(1 + 2 + 3 + \dots + 12) = \frac{78}{12} = \boxed{6.5}.$$

(c)

First,

$$E(X^2) = \frac{1}{12} \sum_{x=1}^{12} x^2 = \frac{650}{12}.$$

The variance is

$$\text{Var}(X) = E(X^2) - [E(X)]^2.$$

Thus,

$$\text{Var}(X) = \frac{650}{12} - (6.5)^2 = \boxed{\frac{143}{12}} \approx \boxed{11.9167}.$$

Q3

Let

$$X \sim N(4, 5).$$

Thus,

$$\mu = 4, \quad \sigma^2 = 5, \quad \sigma = \sqrt{5}.$$

(a)

We want

$$P(X \leq 6).$$

Standardizing,

$$Z = \frac{6 - 4}{\sqrt{5}} = 0.8944.$$

Therefore,

$$P(X \leq 6) = P(Z \leq 0.8944) \approx \boxed{0.8145}.$$

(b)

Since 4 is the mean of the normal distribution,

$$P(X > 4) = \boxed{0.5}.$$

(c)

We want

$$P(-1 \leq X < 1).$$

Standardizing both endpoints,

$$P\left(\frac{-1-4}{\sqrt{5}} \leq Z < \frac{1-4}{\sqrt{5}}\right).$$

Therefore,

$$P(-2.2361 \leq Z < -1.3416) \approx \boxed{0.0772}.$$

Q4

Let

$$Z \sim N(0, 1).$$

(a)

$$P(Z \leq 0.83) = \Phi(0.83) \approx \boxed{0.7967}.$$

(b)

By symmetry,

$$P(-0.83 \leq Z \leq 0.83) = 2\Phi(0.83) - 1.$$

Therefore,

$$P(-0.83 \leq Z \leq 0.83) = 2(0.7967) - 1 \approx \boxed{0.5935}.$$

Q5

Let $X = \{1, 2, \dots, 12\}$ with $f_X(x) = 1/12$.

(a)

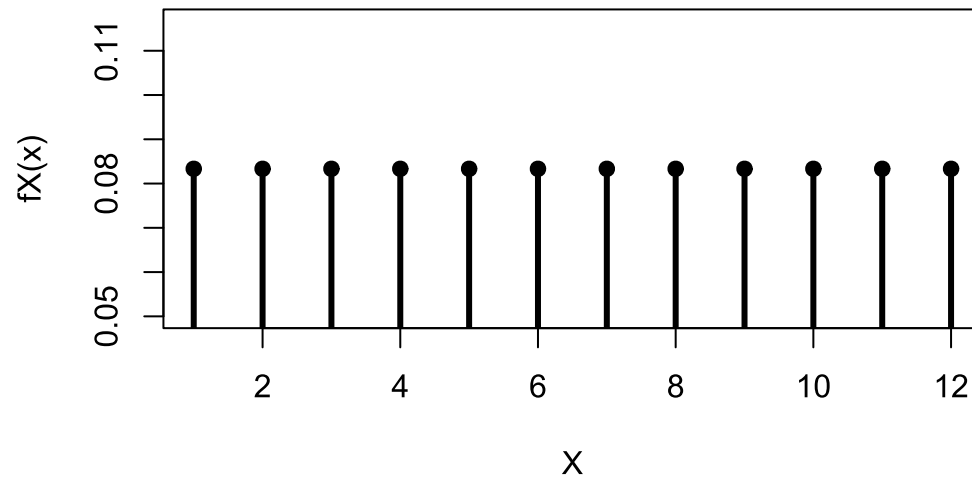
The probability distribution function is plotted below.

```
x <- 1:12
fx <- rep(1/12, 12)

plot(
  x, fx,
  type = "h",
  lwd = 3,
  xlab = "X",
  ylab = "fX(x)",
  main = "Probability Distribution of X"
)

points(x, fx, pch = 19)
```

Probability Distribution of X



(b)

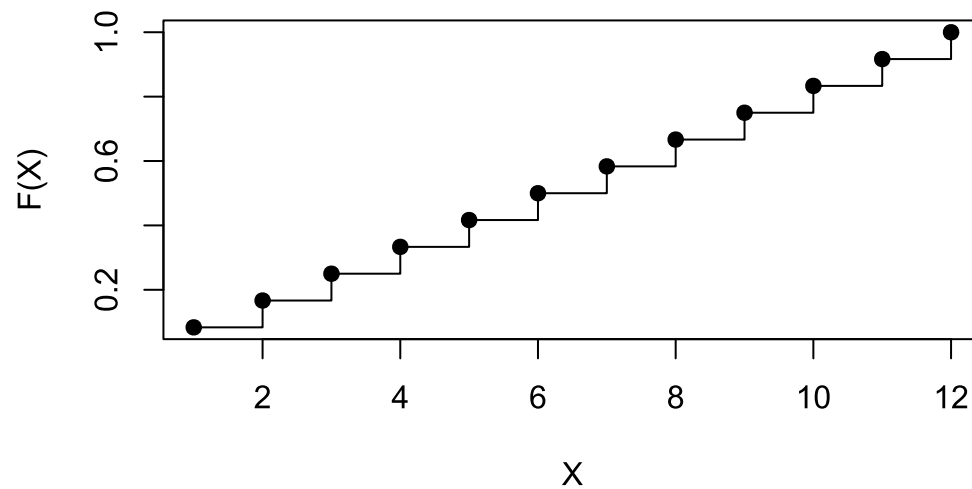
The cumulative probability distribution is:

```
Fx <- cumsum(fx)

plot(
  x, Fx,
  type = "s",
  xlab = "X",
  ylab = "F(X)",
  main = "Cumulative Distribution of X"
)

points(x, Fx, pch = 19)
```

Cumulative Distribution of X



(c)

The expected value is computed in R as follows:

```
EX <- sum(x * fx)
EX
```

```
[1] 6.5
```

Therefore,

$$E(X) = 6.5$$

(d)

The variance is:

```
EX2 <- sum(x^2 * fx)
variance <- EX2 - EX^2
variance
```

```
[1] 11.91667
```

Therefore,

$$\boxed{Var(X) = 11.9167}.$$

Q6

The standard normal density is

$$f(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}.$$

(a)

The function can be defined in R as:

```
f <- function(x) {
  exp(-(x^2)/2) / sqrt(2*pi)
}
```

(b)

Integrating the function over $(-\infty, \infty)$:

```
integrate(f, -Inf, Inf)$value
```

```
[1] 1
```

The result is

$$\boxed{1}.$$

Therefore, the total probability is equal to 1.

(c)

The expected value is:

```
EZ <- integrate(
  function(x) x * f(x),
  -Inf, Inf
)$value
```

```
EZ
```

```
[1] 0
```

Therefore,

$$E(Z) = 0.$$

(d)

The variance is:

```
EZ2 <- integrate(
  function(x) x^2 * f(x),
  -Inf, Inf
)$value

VarZ <- EZ2 - EZ^2
VarZ
```

```
[1] 1
```

Therefore,

$$Var(Z) = 1.$$

(e)

To calculate $P(Z \leq 0.83)$:

```
integrate(f, -Inf, 0.83)$value
```

```
[1] 0.7967306
```

Therefore,

$$P(Z \leq 0.83) \approx 0.7967.$$

(f)

To calculate $P(-0.83 \leq Z \leq 0.83)$:

```
integrate(f, -0.83, 0.83)$value
```

```
[1] 0.5934612
```

Therefore,

$$P(-0.83 \leq Z \leq 0.83) \approx 0.5935.$$

(g)

Using the `pnorm()` command:

```
pnorm(0.83) - pnorm(-0.83)
```

```
[1] 0.5934612
```

The result is approximately

0.5935 .

This is the same result obtained in part (f).

Q7

The probability density function is

$$f(x) = \frac{1}{9}x^2, \quad 0 < x < 3.$$

(a)

Define the function in R:

```
prison_pdf <- function(x) {  
  (1/9) * x^2  
}
```

(b)

Integrate over $[0, 3]$:

```
integrate(prison_pdf, 0, 3)$value
```

```
[1] 1
```

The result is

1 .

Therefore, the density integrates to 1.

(c)

The expected prison sentence is:


```
expected_sentence <- integrate(
  function(x) x * prison_pdf(x),
  0, 3
)$value

expected_sentence
```

```
[1] 2.25
```

Analytically,

$$E(X) = \int_0^3 x \left(\frac{x^2}{9} \right) dx = \frac{1}{9} \int_0^3 x^3 dx.$$

Thus,

$$E(X) = \frac{1}{9} \left[\frac{x^4}{4} \right]_0^3 = \frac{81}{36} = \boxed{2.25}.$$

Therefore, the expected prison sentence is **2.25 years**.

Q8

(a)

The standard normal distribution and Student t distributions are plotted below.

```
curve(
  dnorm(x),
  from = -5,
  to = 5,
  xlim = c(-5, 5),
  ylab = "Density",
  main = "Densities of t Distribution",
  lty = 1
)

curve(
  dt(x, df = 5),
  from = -5,
  to = 5,
  col = 2,
  add = TRUE
```

```

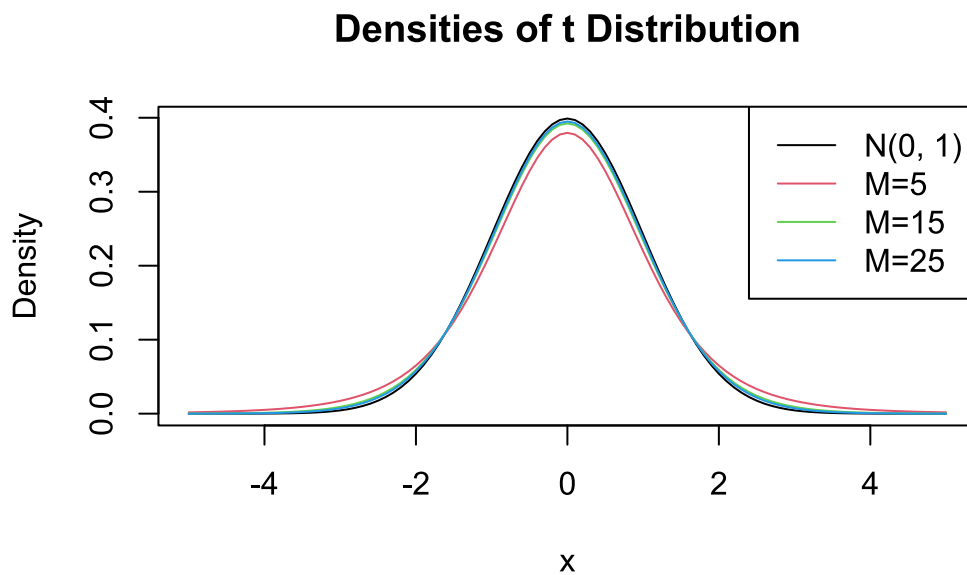
)

curve(
  dt(x, df = 15),
  from = -5,
  to = 5,
  col = 3,
  add = TRUE
)

curve(
  dt(x, df = 25),
  from = -5,
  to = 5,
  col = 4,
  add = TRUE
)

legend(
  "topright",
  c("N(0, 1)", "M=5", "M=15", "M=25"),
  col = 1:4,
  lty = c(1, 1, 1, 1)
)

```



(b)

As the degrees of freedom increase, the Student t distribution becomes closer to the standard normal distribution. With fewer degrees of freedom, the t distribution has heavier tails. As the degrees of freedom increase, the difference between the t distribution and the standard normal distribution becomes smaller.